Note:Students have to write programs along with outputs and get the observation signed by the faculty incharge. For every lab, Observation carries Marks

Solutions

1) Prime Number Check

```
n = int(input("Enter a number: "))

if n <= 1:
    print("Not a prime number")
else:
    count = 0
    i = 2

    while i < n:
        if n % i == 0:
            count = count + 1
        i = i + 1

    if count == 0:
        print("Prime number")
    else:
        print("Not a prime number")
```

2) Factorial Using Loop

```
n = int(input("Enter a number: "))

fact = 1
i = 1

if n < 0:
    print("Factorial is not defined for negative numbers")
else:
    while i <= n:
        fact = fact * i
        i = i + 1

    print("Factorial =", fact)
```

3) Fibonacci Series

```
n = int(input("Enter the number of terms: "))

a = 0
b = 1
i = 1

while i <= n:
    print(a)
    c = a + b
```

```
    a = b
    b = c
    i = i + 1
```

4) Sum of Digits

```
n = int(input("Enter a number: "))

n = abs(n)
sum = 0

while n > 0:
    digit = n % 10
    sum = sum + digit
    n = n // 10

print("Sum of digits =", sum)
```

5) Multiplication Table

```
n = int(input("Enter a number: "))

i = 1

while i <= 10:
    print(n, "x", i, "=", n * i)
    i = i + 1
```

6) Palindrome String Check

```
s = input("Enter a string: ")

reverse = ""

for ch in s:
    reverse = ch + reverse

if s == reverse:
    print("Palindrome")
else:
    print("Not a Palindrome")
```

7) Count Vowels and Consonants

```
s = input("Enter a string: ")

vowels = 0
consonants = 0

for ch in s:
    if ch == 'a' or ch == 'e' or ch == 'i' or ch == 'o' or ch == 'u':
        vowels = vowels + 1
    elif ch == 'A' or ch == 'E' or ch == 'I' or ch == 'O' or ch == 'U':
        vowels = vowels + 1
    elif (ch >= 'a' and ch <= 'z') or (ch >= 'A' and ch <= 'Z'):
        consonants = consonants + 1
```

```
print("Number of vowels:", vowels)
print("Number of consonants:", consonants)
```

8) Reverse a String

```
s = input("Enter a string: ")

reverse = ""

for ch in s:
    reverse = ch + reverse

print("Reversed string:", reverse)
```

9) Anagram Check

```
s1 = input("Enter first string: ")
s2 = input("Enter second string: ")

count_s1 = 0
count_s2 = 0

# Count characters in first string
for ch in s1:
    count_s1 = count_s1 + 1

# Count characters in second string
```

```
for ch in s2:
    count_s2 = count_s2 + 1

if count_s1 != count_s2:
    print("Not an Anagram")
else:
    anagram = True

    for ch in s1:
        count1 = 0
        count2 = 0

        # Count occurrence of ch in first string
        for c in s1:
            if c == ch:
                count1 = count1 + 1

        # Count occurrence of ch in second string
        for c in s2:
            if c == ch:
                count2 = count2 + 1

        if count1 != count2:
            anagram = False
            break

    if anagram:
        print("Anagram")
    else:
```

```
    print("Not an Anagram")
```

10) Count the Number of Digits

```
n = int(input("Enter a positive integer: "))

count = 0

while n > 0:
    n = n // 10
    count = count + 1

print("Number of digits =", count)
```

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